

ACTIVITY 12 — LEARNER WORKSHEET**Value Stream Map and Takt Time**

TYPE	DURATION	DMAIC PHASE
Elective Activity	45 min	MEASURE

Name	
Date	

What you are doing

Extend process mapping into Lean metrics. Map the value stream with process and wait times, calculate process cycle efficiency, then compute takt time to see whether the process can meet customer demand.

What you will produce: A value stream map with process and wait times, a calculated process cycle efficiency and a takt time expressed per unit.

Tools and techniques: SIPOC & Process Map Builder, value stream mapping, lead time, WIP, takt time

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder**data/ — the mock data you analyse:**

- process-steps.csv — 7 rows (step_no, activity, actor, system, process_time_min, wait_time_min, handoff_to)

templates/ — the worksheets you complete:

- takt-time.csv (input, value)
- value-stream-map.csv (step_no, activity, process_time_min, wait_time_min, VA_or_NVA)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Your steps

1. Map the value stream: each step with its process time and the wait time between steps. Start from your Activity 3 process map.

Tool: <https://alfredang.github.io/sipoc/>

2. Total the value-added time and the lead time, then compute process cycle efficiency.
3. Calculate takt time = available working time / customer demand.
4. Compare cycle time against takt time to identify the bottleneck step.

Your working space

Complete every field. Keep this worksheet — it is your reference for applying the method at work.

Step 1 — Map the value stream: each step with its process time and the wait time between steps...

Step 2 — Total the value-added time and the lead time, then compute process cycle efficiency.

Step 3 — Calculate takt time = available working time / customer demand.

Step 4 — Compare cycle time against takt time to identify the bottleneck step.

Answers to the discussion prompts

1. What proportion of total lead time is actually value-added?

2. Which step is the bottleneck against takt time?

3. Where is the biggest single block of waiting, and who owns it?

Reflect & discuss

Estimate the process cycle efficiency of one process you run.

Would your process meet demand if measured against takt time?

Check your work

Your lead time equals the sum of all process and wait times, and takt time is expressed per unit.